

Statistical Learning I

Tutorial: Session 1

July 26, 2026

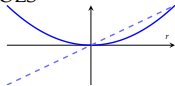
Two loss functions

Model (through the origin): $Y_i = \beta X_i + \varepsilon_i$.

Both OLS and LAD are **M-estimators**: minimize $\sum_i \rho(r_i)$, $r_i = Y_i - \beta X_i$, for a choice of ρ .

OLS

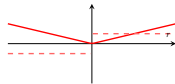
$$\rho_{\text{OLS}}(r) = r^2$$
$$\psi_{\text{OLS}}(r) = \rho'_{\text{OLS}}(r) = 2r \quad \textbf{(unbounded)}$$



LAD

$$\rho_{\text{LAD}}(r) = |r|$$

$$\psi_{\text{LAD}}(r) = \text{sign}(r) \quad \textbf{(bounded)}$$



The seed of everything today

ψ is the influence of one residual on the fit. OLS's influence *grows without bound*; LAD's *saturates at ± 1* . Robustness is entirely a statement about ψ .

Model and estimators (no intercept)

OLS: minimize $\sum_i (Y_i - \beta X_i)^2 \Rightarrow$ closed form

$$\hat{\beta}_{\text{OLS}} = \frac{\sum_i X_i Y_i}{\sum_i X_i^2}.$$

LAD: minimize $\sum_i |Y_i - \beta X_i|$. Write, for $X_i \neq 0$,

$$\sum_i |Y_i - \beta X_i| = \sum_i |X_i| \underbrace{|Y_i/X_i - \beta|}_{r_i}.$$

This is exactly the objective minimized by a **weighted median** of $r_1, \dots, r_n$ with weights $w_i = |X_i|$:

$$\hat{\beta}_{\text{LAD}} = \text{weighted median of } \{Y_i/X_i\} \text{ with weights } \{|X_i|\}.$$

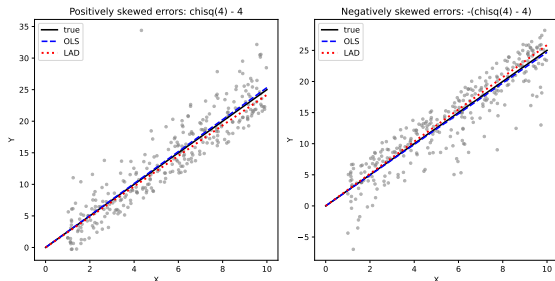
No closed form, but exact and cheap to compute

Sort the ratios r_i , accumulate weight, stop at the first point the cumulative weight reaches 50% of the total. Implemented directly (no numerical optimizer needed) in `lad_origin()` in the R script.

Aside: skewed errors, mean vs. median

Even with *no outliers*, OLS and LAD can disagree if ε is skewed but still mean zero, e.g. $\varepsilon \sim \chi_k^2 - k$.

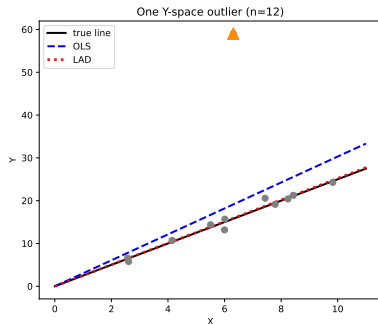
- OLS targets $\mathbb{E}[Y | X] = \beta X$ (exact, since $\mathbb{E}[\varepsilon] = 0$).
- LAD targets $\text{median}(Y | X) = \beta X + \text{median}(\varepsilon)$.
- Positive skew $\Rightarrow \text{median}(\varepsilon) < 0 \Rightarrow$ LAD's line sits *below* OLS's.
- Negate the noise (negative skew) and the shift flips sign.



Mean $\neq$ median for skewed distributions, this is a fundamentally different phenomenon from robustness to contamination.

Y-space outlier: OLS breaks, LAD survives

One observation with an ordinary X but a wildly wrong Y (small n here on purpose, so the effect of the one bad point is unmistakable).



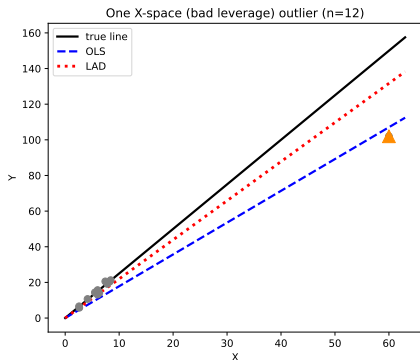
- OLS: squared residual from this point grows quadratically $\Rightarrow$ fit dragged toward it.
- LAD: ψ_{LAD} saturates at ± 1 regardless of how extreme the residual is $\Rightarrow$ fit barely moves.

Ask the class

Can you quantify how much $\hat{\beta}_{\text{OLS}}$ shifts due to *one* observation?

X-space outlier: both break

Now the outlier has an *extreme* X (high leverage), with Y inconsistent with the line.



Discovery question

LAD was robust a moment ago. Why does it fail *here*?

- Weighted-median weights are $w_i = |X_i|$: an extreme X_i carries disproportionate weight and can dominate the median *regardless of residual size*.
- **Key message:** robustness in Y -space and in X -space are *separate* properties. LAD only buys the first.

Adding an intercept

Model (simple linear regression): $Y_i = \alpha + \beta X_i + \varepsilon_i$.

OLS: normal equations, closed form in one line, $\hat{\boldsymbol{\theta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$, $\boldsymbol{\theta} = (\alpha, \beta)^\top$.

LAD: the subgradient (first-order) conditions for $\min_{\alpha, \beta} \sum_i |Y_i - \alpha - \beta X_i|$ are

$$\sum_{i=1}^n \text{sign}(Y_i - \alpha - \beta X_i) = 0, \quad \sum_{i=1}^n X_i \text{sign}(Y_i - \alpha - \beta X_i) = 0.$$

The problem

Two equations, two unknowns, but *non-smooth*. The one-dimensional weighted-median trick does not directly extend to two parameters. This is the entry point for **IRWLS**.

IRWLS: the idea

Simple linear regression, estimated by minimizing the sum of absolute residuals:

$$\min_{\alpha, \beta} \sum_{i=1}^n |Y_i - \alpha - \beta X_i|.$$

IRWLS: the idea

Simple linear regression, estimated by minimizing the sum of absolute residuals:

$$\min_{\alpha, \beta} \sum_{i=1}^n |Y_i - \alpha - \beta X_i|.$$

Key algebraic trick: for any residual $r_i \neq 0$,

$$|r_i| = \frac{r_i^2}{|r_i|}.$$

IRWLS: the idea

Simple linear regression, estimated by minimizing the sum of absolute residuals:

$$\min_{\alpha, \beta} \sum_{i=1}^n |Y_i - \alpha - \beta X_i|.$$

Key algebraic trick: for any residual $r_i \neq 0$,

$$|r_i| = \frac{r_i^2}{|r_i|}.$$

If we *freeze* $w_i = 1/|r_i|$ as fixed weights (evaluated at the *current* (α, β)), then

$$\sum_i |r_i| = \sum_i w_i r_i^2.$$

IRWLS: from trick to algorithm

Minimizing $\sum_i w_i r_i^2$ over (α, β) *with the weights held fixed* is just **weighted least squares**, closed form. Of course the weights depend on the residuals, which depend on (α, β) , so we alternate.

IRWLS: from trick to algorithm

Minimizing $\sum_i w_i r_i^2$ over (α, β) *with the weights held fixed* is just **weighted least squares**, closed form. Of course the weights depend on the residuals, which depend on (α, β) , so we alternate.

Iteratively Reweighted Least Squares (IRWLS), the recipe

- 1 Fix (α, β) , recompute weights from the current residuals.
- 2 Fix the weights, solve weighted least squares for new (α, β) .
- 3 Repeat until (α, β) stops moving.

IRWLS: from trick to algorithm

Minimizing $\sum_i w_i r_i^2$ over (α, β) *with the weights held fixed* is just **weighted least squares**, closed form. Of course the weights depend on the residuals, which depend on (α, β) , so we alternate.

Iteratively Reweighted Least Squares (IRWLS), the recipe

- 1 Fix (α, β) , recompute weights from the current residuals.
- 2 Fix the weights, solve weighted least squares for new (α, β) .
- 3 Repeat until (α, β) stops moving.

Connect the dots

Once IRWLS converges on the Y-space outlier data, which observation ends up with the *smallest* weight?

IRWLS: setting up the algorithm

For the simple linear regression model $Y_i = \alpha + \beta X_i + \varepsilon_i$, write $\mathbf{y} = (Y_1, \dots, Y_n)^\top$, $\mathbf{X}$ the $n \times 2$ design matrix with rows $(1, X_i)$, $\boldsymbol{\theta} = (\alpha, \beta)^\top$.

IRWLS: setting up the algorithm

For the simple linear regression model $Y_i = \alpha + \beta X_i + \varepsilon_i$, write $\mathbf{y} = (Y_1, \dots, Y_n)^\top$, $\mathbf{X}$ the $n \times 2$ design matrix with rows $(1, X_i)$, $\boldsymbol{\theta} = (\alpha, \beta)^\top$.

Step 1: initialize

$$\boldsymbol{\theta}^{(0)} = \hat{\boldsymbol{\theta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

IRWLS: setting up the algorithm

For the simple linear regression model $Y_i = \alpha + \beta X_i + \varepsilon_i$, write $\mathbf{y} = (Y_1, \dots, Y_n)^\top$, $\mathbf{X}$ the $n \times 2$ design matrix with rows $(1, X_i)$, $\boldsymbol{\theta} = (\alpha, \beta)^\top$.

Step 1: initialize

$$\boldsymbol{\theta}^{(0)} = \hat{\boldsymbol{\theta}}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Step 2: residuals from the current fit

$$\mathbf{r}^{(t)} = \mathbf{y} - \mathbf{X}\boldsymbol{\theta}^{(t)}.$$

IRWLS: reweighting

Step 3: reweight

$$w_i^{(t)} = \frac{1}{\max(|r_i^{(t)}|, \delta)}, \quad \mathbf{W}^{(t)} = \text{diag}(w_1^{(t)}, \dots, w_n^{(t)}),$$

for a small numerical floor $\delta > 0$ (guards against $r_i^{(t)} \approx 0$).

IRWLS: reweighting

Step 3: reweight

$$w_i^{(t)} = \frac{1}{\max(|r_i^{(t)}|, \delta)}, \quad \mathbf{W}^{(t)} = \text{diag}(w_1^{(t)}, \dots, w_n^{(t)}),$$

for a small numerical floor $\delta > 0$ (guards against $r_i^{(t)} \approx 0$).

Read this weight carefully

A point with a small residual (fit already passes near it) gets a *large* weight. A point with a large residual (possible outlier) gets a *small* weight. The next step folds these weights into an ordinary weighted least squares solve.

IRWLS: reweighting

Step 3: reweight

$$w_i^{(t)} = \frac{1}{\max(|r_i^{(t)}|, \delta)}, \quad \mathbf{W}^{(t)} = \text{diag}(w_1^{(t)}, \dots, w_n^{(t)}),$$

for a small numerical floor $\delta > 0$ (guards against $r_i^{(t)} \approx 0$).

Read this weight carefully

A point with a small residual (fit already passes near it) gets a *large* weight. A point with a large residual (possible outlier) gets a *small* weight. The next step folds these weights into an ordinary weighted least squares solve.

Quick check

What happens to $w_i^{(t)}$ if $r_i^{(t)} = 0$ exactly?

IRWLS: the algorithm, continued

Step 4: weighted least squares

$$\boldsymbol{\theta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{y}.$$

IRWLS: the algorithm, continued

Step 4: weighted least squares

$$\boldsymbol{\theta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{y}.$$

Stopping rule

Repeat Steps 2–4 until $\|\boldsymbol{\theta}^{(t+1)} - \boldsymbol{\theta}^{(t)}\| < \varepsilon$ for a small tolerance ε .

IRWLS: the algorithm, continued

Step 4: weighted least squares

$$\boldsymbol{\theta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{y}.$$

Stopping rule

Repeat Steps 2–4 until $\|\boldsymbol{\theta}^{(t+1)} - \boldsymbol{\theta}^{(t)}\| < \varepsilon$ for a small tolerance ε .

Why it works

Small residual $\Rightarrow$ large weight (trust this point more). Large residual, i.e. a possible outlier $\Rightarrow$ small weight (down-weighted). This generalizes: for any loss ρ , the weight is $w_i = \rho'(|r_i|)/|r_i|$ (Huber, Tukey bisquare, ...).

IRWLS: the algorithm, continued

Step 4: weighted least squares

$$\boldsymbol{\theta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{y}.$$

Stopping rule

Repeat Steps 2–4 until $\|\boldsymbol{\theta}^{(t+1)} - \boldsymbol{\theta}^{(t)}\| < \varepsilon$ for a small tolerance ε .

Why it works

Small residual $\Rightarrow$ large weight (trust this point more). Large residual, i.e. a possible outlier $\Rightarrow$ small weight (down-weighted). This generalizes: for any loss ρ , the weight is $w_i = \rho'(|r_i|) / |r_i|$ (Huber, Tukey bisquare, ...).

Think about it

If IRWLS kept the Step 1 weights forever instead of recomputing them, would the result still be LAD?

IRWLS: step-by-step trace

A small example, $n = 12$, true $(\alpha, \beta) = (3, 2.5)$:

t	$\alpha^{(t)}$	$\beta^{(t)}$	$\ \boldsymbol{\theta}^{(t)} - \boldsymbol{\theta}^{(t-1)}\ $
0 (OLS init)	3.6429	2.4660	—
1	3.2393	2.5260	4.08×10^{-1}
2	3.2069	2.5329	3.31×10^{-2}
5	3.0855	2.5486	4.50×10^{-2}
10	2.9117	2.5684	1.96×10^{-2}
15	2.8904	2.5709	4.60×10^{-4}
20	2.8902	2.5709	4.88×10^{-6}
24 (converged)	2.8901	2.5709	4.18×10^{-13}

- Monotone-looking approach to a fixed point; $\|\Delta\boldsymbol{\theta}\|$ shrinks roughly geometrically once close to the solution.
- Exact numbers will differ slightly when you run the R script (different random seed), but the convergence pattern is the same.

An alternate algorithm: alternating conditional medians

Both conditional problems have an *exact* closed form:

- Fix β : $\min_{\alpha} \sum_i |(Y_i - \beta X_i) - \alpha| \Rightarrow \alpha = \text{median}(Y_i - \beta X_i)$.
- Fix α : $\min_{\beta} \sum_i |(Y_i - \alpha) - \beta X_i| \Rightarrow \beta = \text{weighted median of } \frac{Y_i - \alpha}{X_i}$ with weights $|X_i|$ (Section 3's trick, applied to $Y_i - \alpha$).

The natural idea

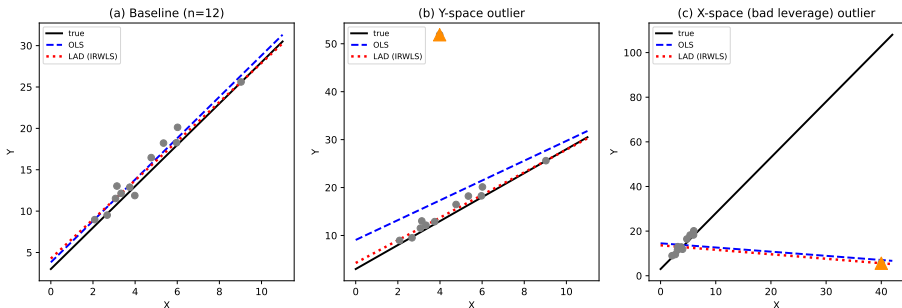
Alternate: fix $\beta^{(t)}$, get $\alpha^{(t+1)}$ by an ordinary median; fix $\alpha^{(t+1)}$, get $\beta^{(t+1)}$ by a weighted median; repeat until both stop moving. Each half-step is an *exact* minimization, cheap, no linear solve needed.

Ask the class

Does alternating two exact minimizations like this have to converge to the true joint LAD minimizer?

Outlier experiments revisited (intercept model)

Same two scenarios, now with $(\hat{\alpha}, \hat{\beta})$ from OLS and from LAD via IRWLS:



Same conclusions as the no-intercept model, now you have watched the algorithm arrive there.

Open questions

The IRWLS weight does not see leverage

$w_i = 1/|r_i|$ downweights *large residuals*. But a high-leverage point can have a *small* residual and still be highly influential (it can pull the whole line toward itself, leaving its own residual small). So IRWLS does **not** fix the X-space problem either.

Question to leave open

What would a weight function that accounts for leverage look like?

Assignment (see R script, Step 14)

What happens if $\varepsilon \sim \text{Cauchy}$?

Open questions: one more direction

Quantile regression

LAD is the $\tau = 0.5$ case of a more general family: pick $\rho_\tau(r) = r(\tau - \mathbb{1}(r < 0))$ and you target the τ -th conditional quantile of $Y \mid X$ instead of the median. Same M -estimator machinery, same IRWLS-style fitting idea, different ρ . Pointer for later, not today's topic.